

# Do LLM fingerprints survive outside English?

Interpretable attribution of five frontier models across seven languages, under translation, and against the models themselves as judges

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59%–72%

five-way attribution accuracy from 21 interpretable features, in every one of 7 languages (chance 20%)

p = 0.36

no resource gradient: accuracy does not fall from English to Hindi (cell-level GLM, n = 839)

42/42

cross-lingual transfers above chance: the fingerprint is largely language-invariant (mean 0.49)

ρ = -0.96

models converge stylistically as resources fall: separation shrinks monotonically to Hindi

62%

attribution after Google Translate; English-trained classifier reads translations at 66%

25% vs 20%

only Claude beats chance as a judge of authorship; the other four default to one fixed answer

## Why

Provenance tools are built and tested in English, but the need to attribute text to a specific model — disinformation tracing, academic integrity, moderation — is mostly non-English. Detection is known to weaken off English; attribution to a *specific* model has been English-only and black-box.

Nobody had measured attribution with **interpretable** features across a resource gradient, separated model style from language style, tested convergence in low-resource languages, or checked whether translation launders the signal.

## Design

**5 models:** GPT-5.5 · Gemini 3.5 Flash · Claude Opus 4.7 · Grok 4.3 · DeepSeek V4 Pro (one OpenRouter key; served model strings logged).

**7 languages** in decreasing training-data share: English → Spanish → Chinese → Russian → Japanese → Turkish → Hindi.

**12 essay prompts**, each fixing a topic, a stance (6 for / 6 against) and three supporting points. A non-subject model (Llama 4) writes a native version of each prompt in every language — never a translation — and a third-family reviewer (Qwen) verifies every version against the original. Content is held fixed; only style is free.

**840 responses** (× 2 generations), no system prompt, temperature 0.7, reasoning traces excluded. 839 pass language ID, length and refusal checks.

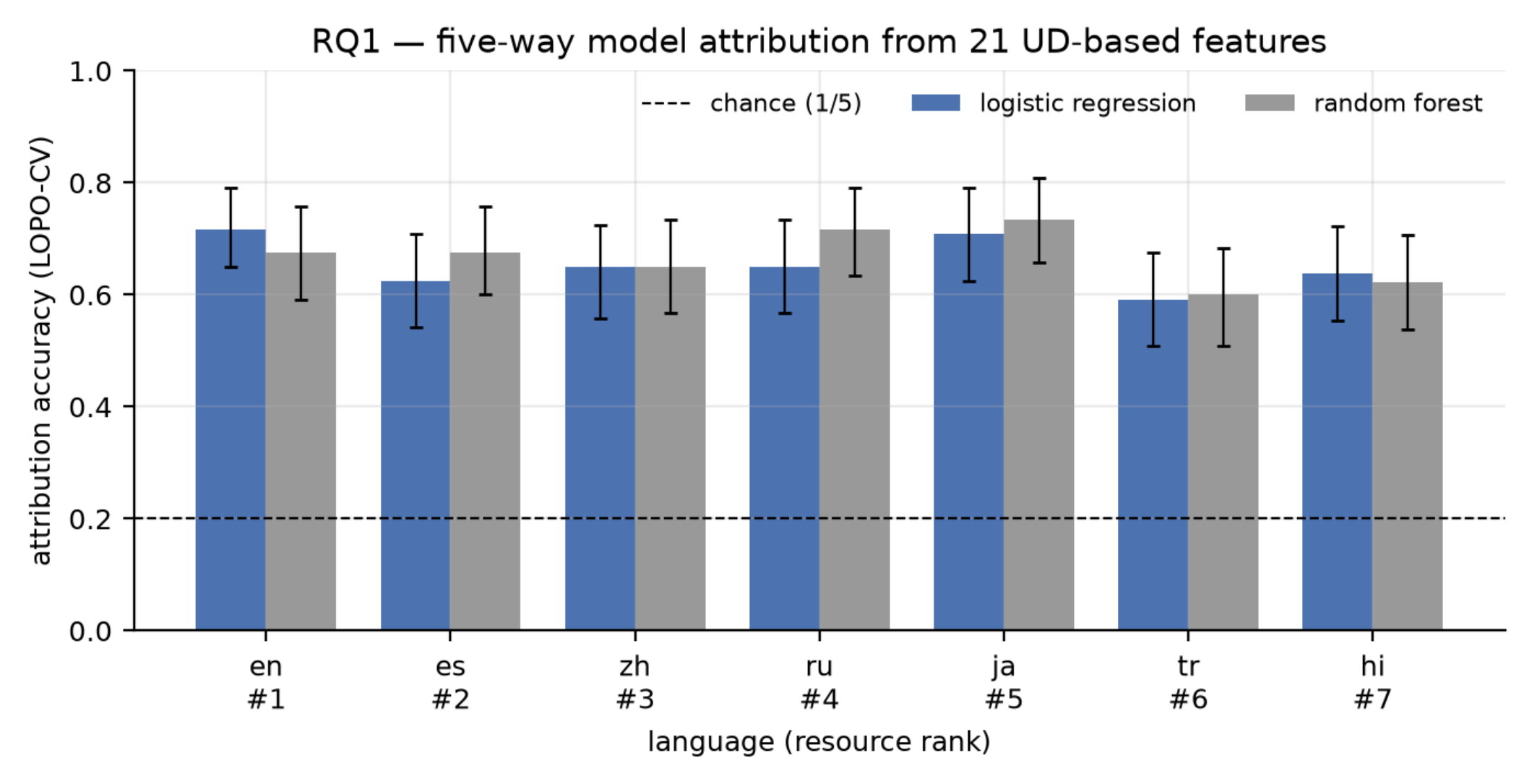
## 21 features, one definition in every language

All text parsed with Stanza on Universal Dependencies, so each feature means the same thing in Chinese, Turkish or Hindi.

**Lexical** MATTR · hapax rate · token length · Zipf slope  
**Syntactic** sentence length  $\mu$ ,  $\sigma$  · burstiness · dependency depth · subordination · function-word ratio · first person  
**Structure** paragraph count, length · question rate · connectives  
**Punctuation** comma · colon · dash · semicolon per 1k, script equivalents mapped  
**Character** bigram entropy · digit rate

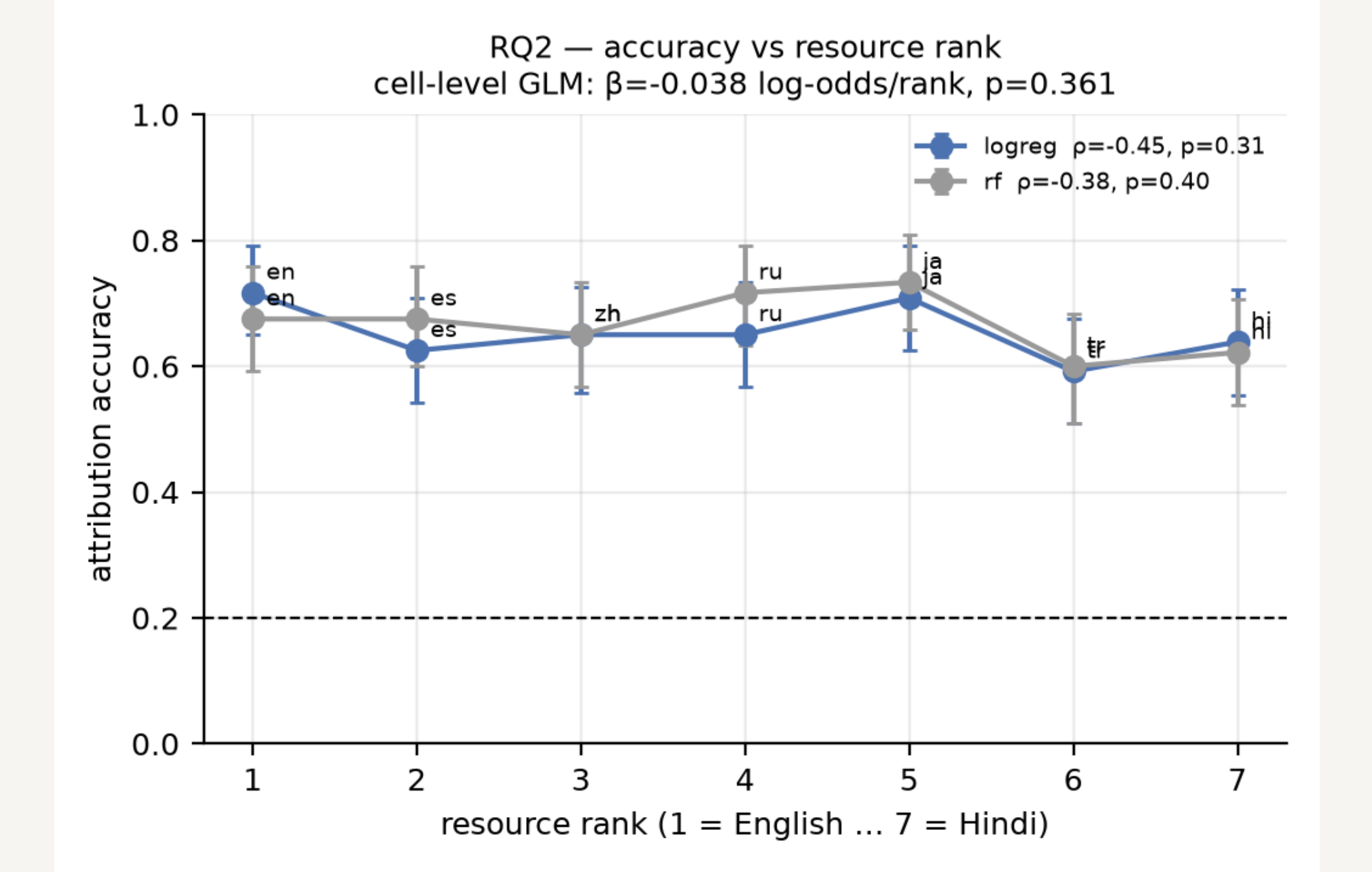
English-only measures (Flesch, Fog, hedges, passive, contractions) are deliberately excluded. All tests pre-specified before collection.

## 1 · Attribution works in every language



Leave-one-prompt-out CV; logistic regression on standardised features (interpretable) and random forest (ceiling). Every CI excludes chance. GPT and Grok are most identifiable; DeepSeek the blur.

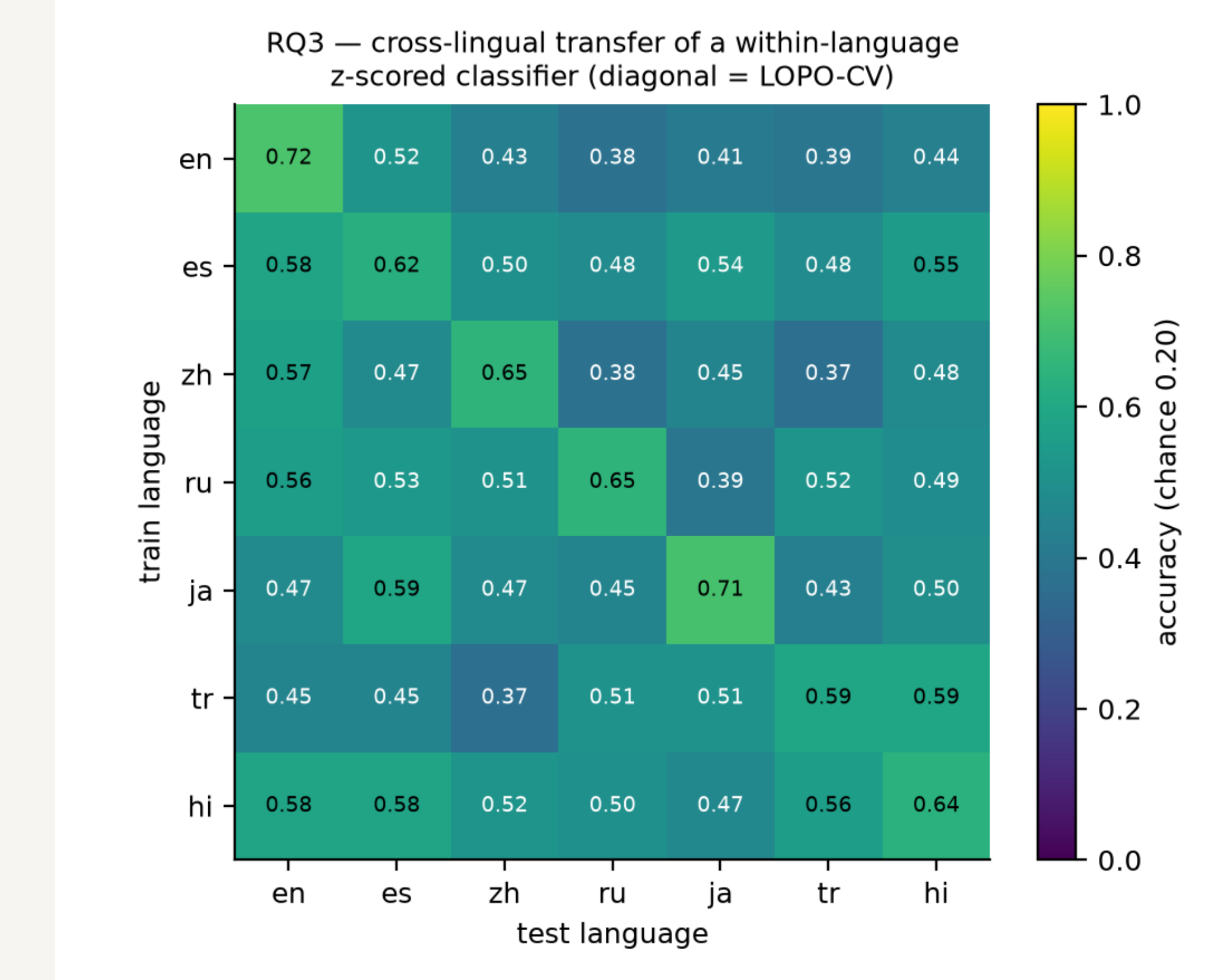
## 2 · ... and does not fade with resource level



Spearman  $\rho = -0.45$  ( $p = 0.31$ ); cell-level GLM  $\beta = -0.038$  log-odds per rank ( $p = 0.36$ ). Japanese matches English; Hindi beats Spanish. The pre-registered gradient hypothesis is **not** supported.

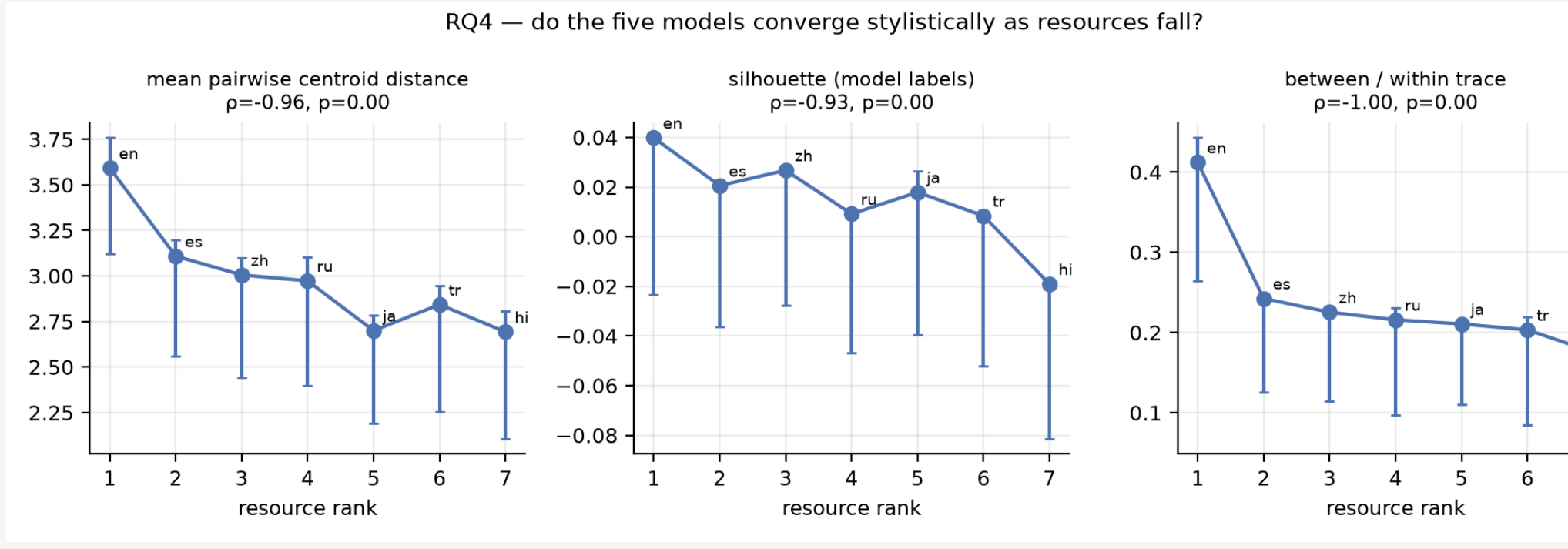
The fingerprint is structural: paragraph count, comma and colon density, and lexical-diversity slope carry it; dependency depth and subordination barely contribute.

## 3 · Model style is mostly language-invariant



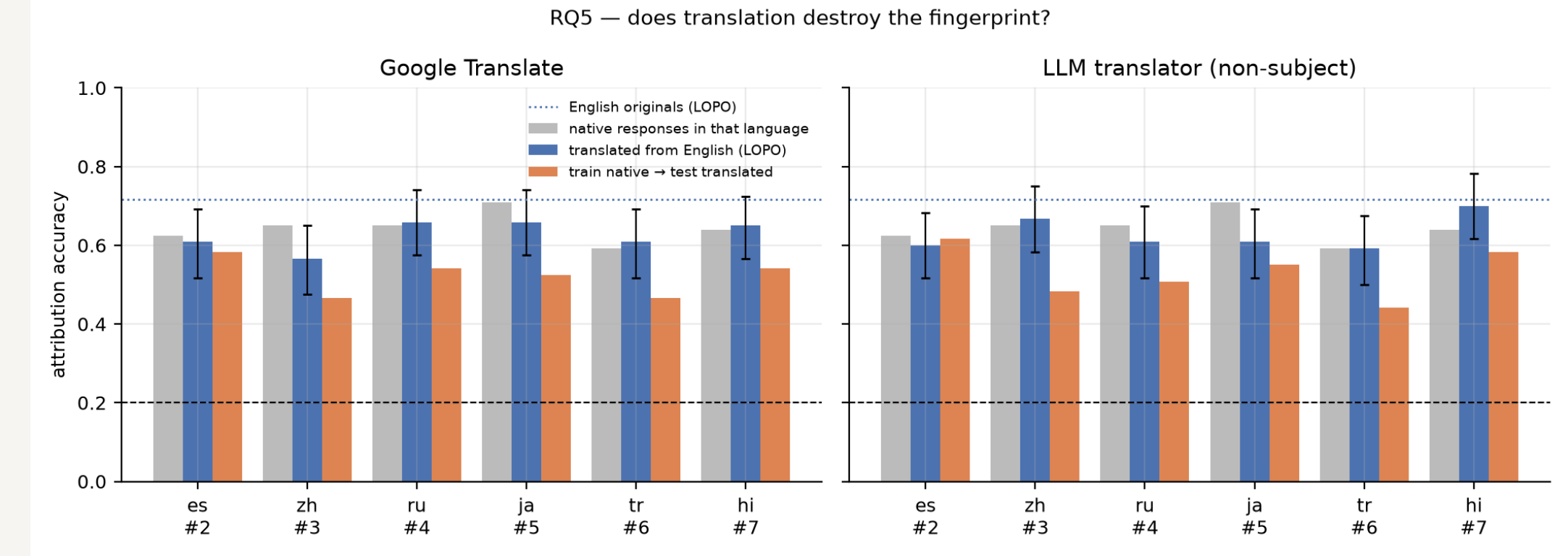
Language explains most feature variance (mean partial  $\eta^2$  0.54 vs 0.14 for model, 0.13 interaction). But remove each language's mean and a classifier trained on any language beats chance on every other: 42/42 pairs, mean 0.49 vs 0.65 within-language. About three quarters of the fingerprint transfers.

## 4 · Yet models converge as resources fall



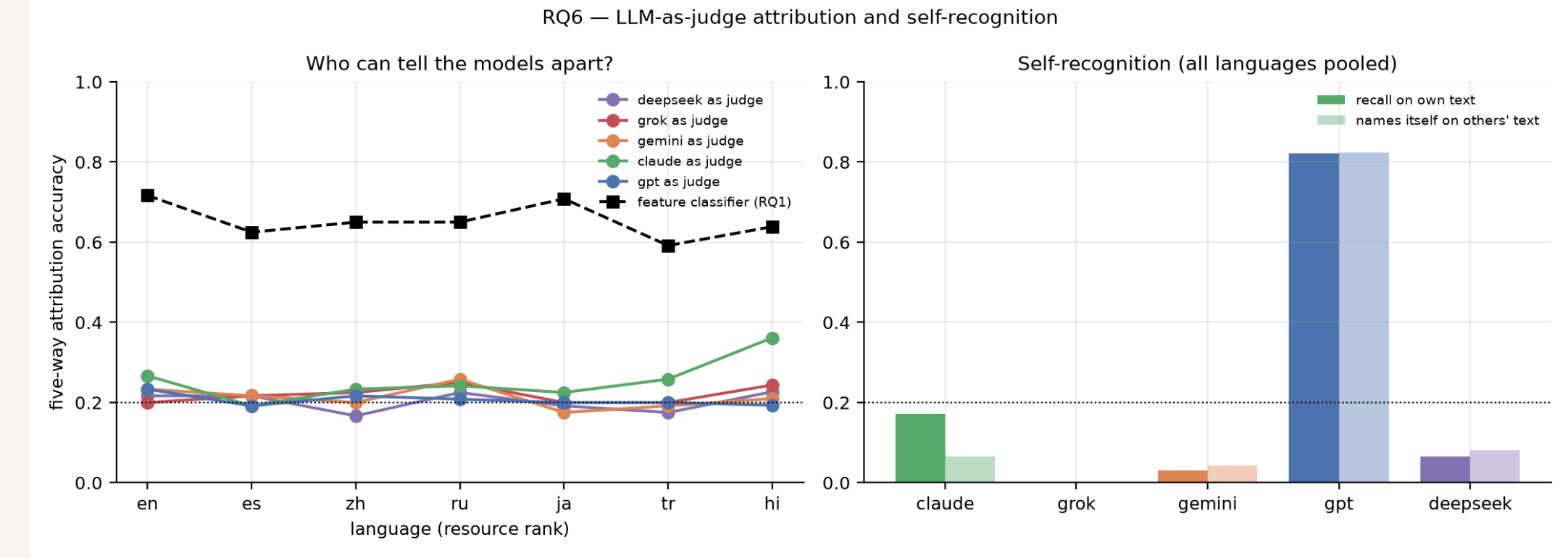
Within-language separation of the five model centroids falls monotonically from English to Hindi (centroid distance  $\rho = -0.96$ ; between/within ratio 0.41 → 0.18). Almost all of it is Grok's terse English register disappearing into the pack. Convergence in the bulk of features, attribution sustained by a few.

## 5 · Translation does not launder the fingerprint



All 120 English responses translated into the six other languages by Google Translate and by a free open model (MiniMax M3). Attribution on translated text alone: 0.62 / 0.63. An **English-trained** classifier reads the translations at 0.66: structure (paragraph count  $\rho = 1.00$ , sentence length 0.89) survives; vocabulary is rewritten (MATTR 0.3).

## 6 · Baseline: the models themselves cannot do this



Each model asked which of the five wrote each text (reasoning off, options shuffled). GPT 20.6%, Grok 21.9%, Gemini 21.2%, DeepSeek 20.3%: chance, each defaulting to one answer (GPT names itself 82% of the time). Claude 25.4% is the only judge above chance, and only just. The fingerprint is real but not one the models read off their own output.

## Takeaways

- Interpretable, UD-based features attribute LLM text in every language and beat every LLM judge by 35–50 points.
- Fingerprints do not fade with resource level and survive translation — provenance across a language boundary works from an English-trained model.
- Low-resource registers are flattening: the five models write more alike in Hindi than in English.
- Limits: one genre; length is part of the linear signal; ordinal resource rank; dated model snapshots.